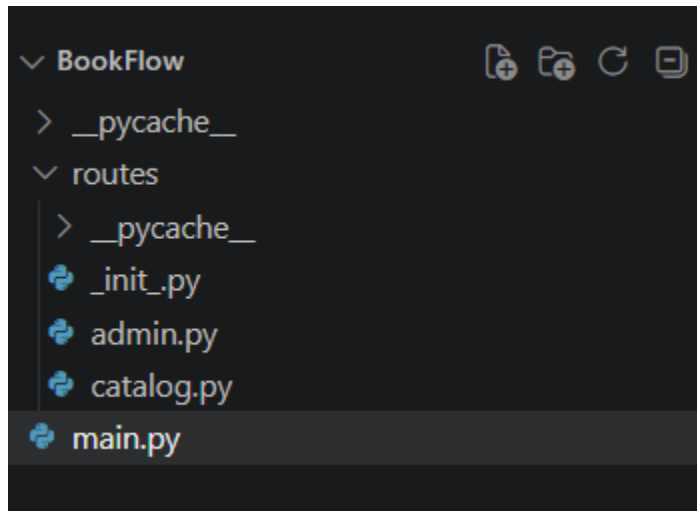


PRACTICAL WEEK - 6 : BookFlow – Organized Routing & CRUD

1. Project Structure :



2. catalog.py:

```
routes > catalog.py > get_book
1  from fastapi import APIRouter, HTTPException
2
3  router = APIRouter()
4
5  # Temporary book data
6  books = [
7      {
8          "id": 101,
9          "title": "Python Programming",
10         "author": "John Smith",
11         "price": 450,
12         "isbn": "9781234567890",
13         "status": "Available"
14     },
15     {
16         "id": 102,
17         "title": "Database Management",
18         "author": "Robert Brown",
19         "price": 550,
20         "isbn": "9789876543210",
21         "status": "Available"
22     }
23 ]
24
25 # GET /books
26 @router.get("/books")
27 def get_books():
28     return books
29
30 # GET /books/{book_id}
31 @router.get("/books/{book_id}")
32 def get_book(book_id: int):
33     for book in books:
34         if book["id"] == book_id:
35             return book
36
37     raise HTTPException(
38         status_code=404,
39         detail="Book not found"
40 )
```

3. admin.py:

```
routes > admin.py > delete_book

1  from fastapi import APIRouter, HTTPException
2  from pydantic import BaseModel, Field
3  from routes.catalog import books
4
5  router = APIRouter()
6
7  # Pydantic model
8  class Book(BaseModel):
9      title: str
10     author: str
11     price: float = Field(ge=0)
12     isbn: str
13     status: str
14
15     # POST /admin/books
16     @router.post("/admin/books")
17     def add_book(book: Book):
18         new_id = max([b["id"] for b in books], default=100) + 1
19         new_book = {
20             "id": new_id,
21             **book.model_dump()
22         }
23         books.append(new_book)
24         return {
25             "message": "Book added successfully",
26             "book": new_book
27         }
28
29     # PUT /admin/books/{book_id}
30     @router.put("/admin/books/{book_id}")
31     def update_book(book_id: int, book: Book):
32         for index, existing_book in enumerate(books):
33             if existing_book["id"] == book_id:
34                 updated_book = {
35                     "id": book_id,
36                     **book.model_dump()
37                 }
38                 books[index] = updated_book
39                 return {
40                     "message": "Book updated successfully",
41                     "book": updated_book
42                 }
43             raise HTTPException(
44                 status_code=404,
45                 detail="Book not found"
46             )
47
48     # DELETE /admin/books/{book_id}
49     @router.delete("/admin/books/{book_id}")
50     def delete_book(book_id: int):
51         for index, book in enumerate(books):
52             if book["id"] == book_id:
53                 deleted_book = books.pop(index)
54                 return {
55                     "message": "Book deleted successfully",
56                     "book": deleted_book
57                 }
58             raise HTTPException(
59                 status_code=404,
60                 detail="Book not found"
61             )
```

4. main.py :

```
main.py > ...
1  from fastapi import FastAPI
2  from routes.catalog import router as catalog_router
3  from routes.admin import router as admin_router
4
5  app = FastAPI(
6      |   title="BookFlow Library API"
7  )
8
9  app.include_router(catalog_router)
10 app.include_router(admin_router)
```

5. Swagger UI :

BookFlow Library API 0.1.0 OAS 3.1

/openapi.json

default

GET	/books	Get Books	▼
GET	/books/{book_id}	Get Book	▼
POST	/admin/books	Add Book	▼
PUT	/admin/books/{book_id}	Update Book	▼
DELETE	/admin/books/{book_id}	Delete Book	▼

Schemas

Book > Expand all object

HTTPValidationError > Expand all object

ValidationError > Expand all object

6. GET books :

GET

/books

Get Books

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Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books
```

Server response

Code

Details

200

Response body

```
[
  {
    "id": 101,
    "title": "Python Programming",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  },
  {
    "id": 102,
    "title": "Database Management",
    "author": "Robert Brown",
    "price": 550,
    "isbn": "9989876543210",
    "status": "Available"
  }
]
```

Download

Response headers

```
content-length: 240
content-type: application/json
date: Wed, 16 Sep 2026 09:16:45 GMT
server: uvicorn
```

7.GET books by id :

GET

/books/{book_id}

Get Book

^

Parameters

Cancel

Name	Description
book_id * required	
integer	101
(path)	

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/101' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/101
```

Server response

Code	Details
200	Response body<pre>{ "id": 101, "title": "python Programming", "author": "John Smith", "price": 450, "isbn": "9781234567890", "status": "Available" }</pre> Response headers<pre>content-length: 117 content-type: application/json date: Wed, 16 Sep 2026 09:17:44 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

8. POST books :

POST/admin/booksAdd Book

Parameters

CancelReset

No parameters

Request bodyrequired

application/json

Edit ValueSchema

```
{
  "title": "Artificial Intelligence",
  "author": "David Miller",
  "price": 600,
  "isbn": "9781122334455",
  "status": "Available"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/admin/books' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "title": "Artificial Intelligence",
  "author": "David Miller",
  "price": 600,
  "isbn": "9781122334455",
  "status": "Available"
}'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

CodeDetails

200

Response body

```
{
  "message": "Book added successfully",
  "book": {
    "id": 103,
    "title": "Artificial Intelligence",
    "author": "David Miller",
    "price": 600,
    "isbn": "9781122334455",
    "status": "Available"
  }
}
```

Download

Response headers

```
content-length: 171
content-type: application/json
date: Wed, 16 Sep 2026 09:19:13 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

9. PUT books by ID :

PUT

/admin/books/{book_id} Update Book

Parameters

Cancel

Reset

Name	Description
book_id * required integer (path)	103

Request body required

application/json

Edit Value | Schema

```
{
  "title": "Advanced Artificial Intelligence",
  "author": "David Miller",
  "price": 700,
  "isbn": "9781122334455",
  "status": "Available"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/admin/books/103' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "title": "Advanced Artificial Intelligence",
    "author": "David Miller",
    "price": 700,
    "isbn": "9781122334455",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books/103

Server response

Code	Details
200	Response body <pre>{ "message": "Book updated successfully", "book": { "id": 103, "title": "Advanced Artificial Intelligence", "author": "David Miller", "price": 700, "isbn": "9781122334455", "status": "Available" } }</pre> Download Response headers <pre>content-length: 182 content-type: application/json date: Wed, 16 Sep 2026 09:21:25 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links
	Media type application/json Controls Accept header. Example Value Schema <pre>"string"</pre>	
422	Validation Error	No links

10 . DELETE books by ID:

DELETE

/admin/books/{book_id} Delete Book

⌵

Parameters

Cancel

Name	Description
book_id required	
integer	103
(path)	

ExecuteClear

Responses

Curl

```
curl -X 'DELETE' \
'http://127.0.0.1:8000/admin/books/103' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/admin/books/103
```

Server response

Code	Details
200	Response body <pre>{ "message": "Book deleted successfully", "book": { "id": 103, "title": "Advanced Artificial Intelligence", "author": "David Miller", "price": 700, "isbn": "9781122334455", "status": "Available" } }</pre> Download Response headers <pre>content-length: 182 content-type: application/json date: Wed, 16 Sep 2026 09:22:57 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links